

Compact VCSEL–Metasurface OAM Transmitter with SIMO Processing for Optical Wireless Interconnects

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Abstract *Optical wireless interconnects require compact transmitters and robust detection. We demonstrate a VCSEL–metasurface OAM link with SIMO processing for short-reach interconnects. At the HD-FEC threshold, ZF achieves 3.84 Gbps (no turbulence) and 1.96 Gbps ($D/r_0 = 0.2$), outperforming MRC, confirming interference-limited operation under the considered receiver configuration.*

Introduction

The continued scaling of cloud computing and artificial intelligence (AI) workloads is driving demand for high-density, energy-efficient optical wireless interconnects in data centers, motivating free-space links as a flexible alternative to fiber. Orbital angular momentum (OAM) beams provide an additional spatial degree of freedom for multiplexing and diversity [1]. In intra-data-center environments, weak but structured perturbations, such as airflow and thermal gradients, can induce intermodal mixing and correlated distortions even over short distances [2, 3], limiting practical deployment due to the lack of compact transmitter architectures and the sensitivity of structured-light links to modal crosstalk.

Most OAM-based free-space demonstrations rely on bulk optics, spatial light modulators (SLM), and high-power laser sources, often combined with coherent detection and multiple-input multiple-output (MIMO) digital signal processing (DSP) to mitigate turbulence-induced impairments and intermodal mixing [3–5]. While these approaches achieve high capacity and strong turbulence resilience, they incur significant complexity, power consumption, and alignment overhead, making them unsuitable for short-reach intra-data-center interconnects. Modal diversity has been proposed to improve robustness by transmitting redundant data across multiple spatial modes [6].

Recent advances in metasurface optics enable compact, planar devices capable of simultaneous beam shaping and collimation, providing a pathway toward integrated OAM transmitters. However, most metasurface-based demonstrations focus on beam generation or multiplexing in fiber or free-space systems without diversity-receiver processing [7–10]. While [11] demonstrates OAM-based diversity gain using metasurface-generated modes and combining techniques, it relies on high-power sources and external collimation, limiting applicability to compact, low-size, weight, and power (SWaP) systems.

In this paper, we experimentally demonstrate a compact vertical cavity surface emitting laser (VCSEL)–metasurface OAM transmitter for short-reach optical wireless interconnects, operating in an intensity modulation and direct detection (IM-DD) configuration without external collimation optics. A dielectric metasurface simultaneously collimates the beam and generates conjugate OAM modes ($\ell = \pm 1$) carrying identical data, forming two diversity branches. The use of OAM-based structured light enables compact generation of spatial diversity branches without additional optical components. At the receiver, SLM-based mode decomposition [12, 13] enables an effective 1×2 single input-multiple output (SIMO) system, where imperfect mode decomposition produces correlated and power-imbalanced observations across the two receiver branches. Unlike prior OAM multiplexing demonstrations (see Table 1), this work focuses on diversity-based reception using structured light, combining compact VCSEL–metasurface integration, low-complexity IM-DD, and interference-aware SIMO processing to recover performance under practical receiver non-idealities, providing a pathway toward low-power, chip-scale optical interconnects.

Experimental Setup and Methodology

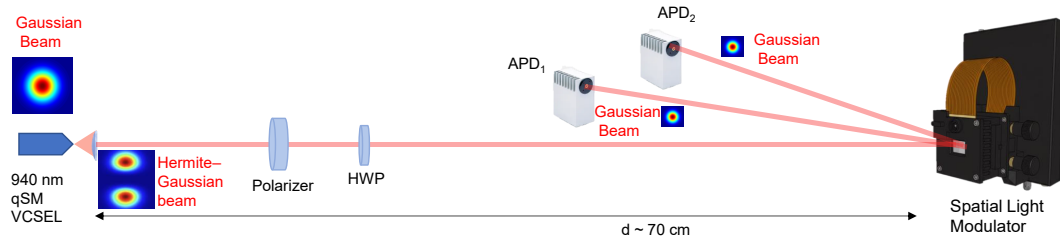
The experimental setup of the compact OAM-enabled optical wireless link is shown in Fig. 1. A 940 nm VCSEL (VISystem qSM-940M) is directly modulated using an optical orthogonal frequency division multiplexing (optical-OFDM) signal generated by an arbitrary waveform generator (AWG) (Keysight M8195A, 1 V_{pp}) and combined with a direct current (DC) bias (2 V, 3 mA, Keysight E36313A) via a bias-tee (Mini-Circuits ZFBT-4R2GW+), yielding ~ 0.5 mW optical output.

A dielectric metasurface placed at the VCSEL output performs simultaneous beam collimation and modal encoding, converting the emitted Gaussian beam into an HG_{10} mode (superposition of $\ell = \pm 1$ OAM states) with beam size $D = 0.5$ mm, eliminating bulk optics and enabling a compact

Tab. 1: Selected prior work positioning the proposed compact VCSEL–metasurface OAM link.

Ref	Tx	*Comp.	Regime	Processing	Limitation vs. this work
[4]	Transponder and lantern	No	Strong turb.	SIC-MIMO	Coherent, EDFA, SLM; high complexity.
[5]	SLM	No	Strong turb.	Adaptive MIMO	Coherent; bulky optics.
[6]	HG/LG modes	No	Turbulence	Diversity	No OAM Tx integration; not IM/DD.
[14]	Bulk optics	No	Dynamic turb.	Pre-comp.	Bulk optics; high complexity Tx.
[7–10]	Metasurface	Yes	Fiber/ FSO	None	No diversity processing.
[11]	Metasurface	No	Turbulence	MRC	External optics at Tx, high-power source; requires EDFA, not compact IM/DD.
This work	VCSEL and metasurface	Yes	Weak	ZF/MRC (SIMO)	Compact IM/DD; no external Tx. optics; low-SWaP.

Note: *Comp.: compact transmitter; Yes indicates no external collimation optics or bulk components beyond the VCSEL and metasurface. IM/DD: intensity modulation/direct detection; EDFA: erbium-doped fiber amplifier; SLM: spatial light modulator; MRC: maximum ratio combining; ZF: zero forcing; SIMO: single-input multiple-output.

**Fig. 1:** Experimental setup of the compact VCSEL–metasurface SIMO optical wireless link.

structured-light transmitter.

The beam propagates over ~ 70 cm. Channel impairments are emulated using a phase-only SLM (Meadowlark Optics S-19.12) applying a composite phase mask for turbulence and controlled modal separation [13]. Turbulence is modeled using Kolmogorov phase screens parameterized by D/r_0 .

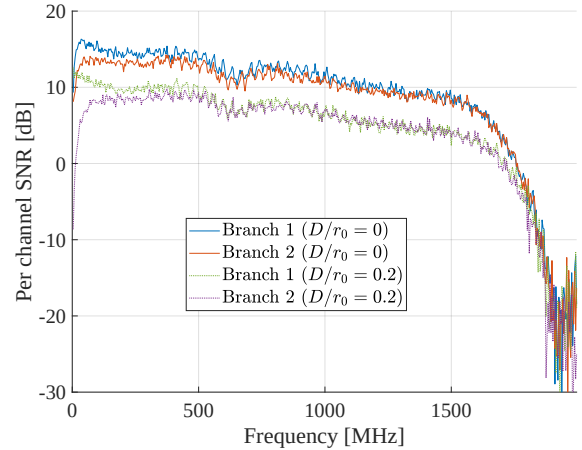
The resulting spatial components are detected at 30 cm by two avalanche photo detectors (APDs) (Hamamatsu C5658, 3 dB bandwidth = 1 GHz), forming the two receiver branches of a 1×2 SIMO system. The electrical signals are captured by a 10 GHz oscilloscope (Keysight UXR0104B) and processed offline in MATLAB®. The system employs DC-biased optical orthogonal frequency division multiplexing (DCO-OFDM) with quadrature amplitude modulation (QAM) modulation, using pseudorandom bit sequences and Hughes–Hartogs bit and power loading; OFDM parameters follow [15].

The system is modeled as a 1×2 SIMO channel with inter-branch mixing:

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}, \quad (1)$$

where $\mathbf{H}$ captures inter-branch mixing, branch correlation, and power imbalance between the two receiver branches due to imperfect mode decomposition and turbulence.

Three receiver strategies are evaluated: no processing, maximal ratio combining (MRC) for diversity combining, and zero forcing (ZF) for

**Fig. 2:** Measured per-branch SNR as a function of frequency for both receiver branches, under no turbulence and turbulence ($D/r_0 = 0.2$).

interference-aware SIMO combining to mitigate inter-branch mixing and residual intermodal mixing.

Results and Discussion

The link is characterized by per-branch and post-processing signal-to-noise ratio (SNR), shown in Fig. 2 and Fig. 3. Without turbulence, the two receiver branches exhibit a low-frequency SNR asymmetry of ≈ 2.19 dB, indicating residual imbalance in the transmitter–receiver path and imperfect mode decomposition. Both branches follow a bandwidth-limited roll-off, approaching 0 dB around 1.6 GHz due to APD bandwidth.

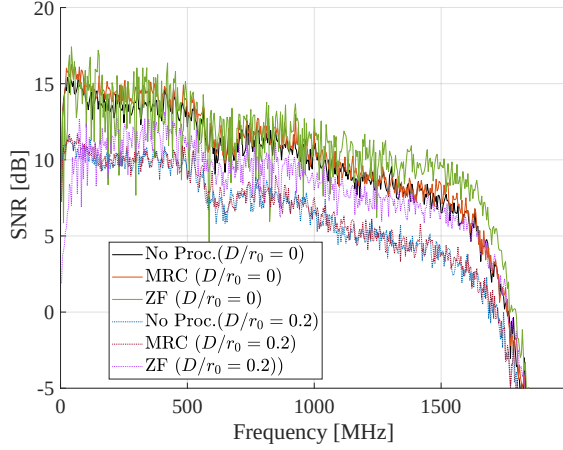


Fig. 3: Post-processing SNR as a function of frequency for no processing, MRC, and ZF combining, under no turbulence and weak turbulence ($D/r_0 = 0.2$).

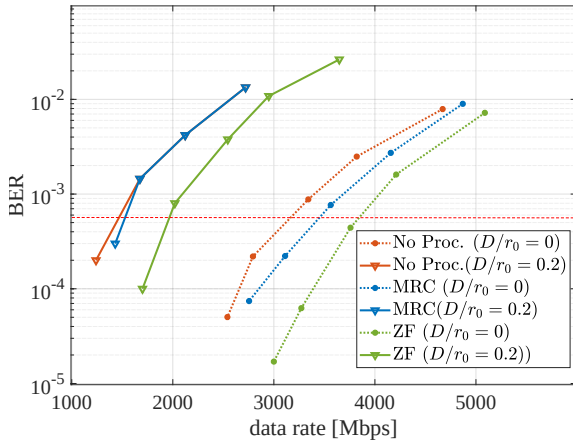


Fig. 4: Measured BER as a function of data rate for no processing, MRC, and ZF combining, under no turbulence and weak turbulence ($D/r_0 = 0.2$). The dashed line indicates the 7% HD-FEC threshold at $\text{BER} = 3.8 \times 10^{-3}$.

Under turbulence ($D/r_0 = 0.2$), an overall SNR degradation of ≈ 5 dB is observed while preserving the roll-off trend, indicating that system response remains bandwidth-limited, with turbulence introducing an additional power penalty together with increased branch correlation and inter-branch mixing.

Post-processing SNR shows a frequency-dependent trade-off between MRC and ZF. At low frequencies, where branch imbalance dominates, ZF suffers from noise enhancement due to decoupling, and MRC achieves higher SNR through power combining. As frequency increases and branch SNRs become comparable, the dominant impairment shifts to branch correlation and inter-branch mixing. In this regime, ZF mitigates interference between the two correlated receiver branches, arising from imperfect mode decomposition, achieving an SNR advantage over MRC that translates to improved BER and higher data rates.

The BER versus data rate is shown in Fig. 4, with the 7% Hard-Decision Forward Error Correction (HD-FEC) threshold at 3.8×10^{-3} . Without turbulence, ZF achieves 3.84 Gbps, outperforming MRC (3.49 Gbps) and no processing (3.21 Gbps), indicating that residual inter-branch mixing, modal crosstalk, and branch imbalance limit performance even in static conditions and can be mitigated through linear equalization.

Under turbulence ($D/r_0 = 0.2$), all schemes degrade, particularly at higher rates. At the FEC threshold, ZF achieves 1.96 Gbps, outperforming MRC (1.58 Gbps) and no processing (1.49 Gbps), with the relative ordering preserved. This indicates that turbulence does not reduce the channel to purely uncorrelated fading, but maintains structured correlation and inter-branch mixing, allowing interference-aware processing such as ZF to remain effective.

The throughput advantage of ZF increases under turbulence, consistent with improved suppression of inter-branch mixing in the bandwidth-limited regime. Overall, the system operates in an interference-dominated regime, where branch correlation, inter-branch mixing, imperfect mode decomposition, and residual modal crosstalk—arising from hardware imperfections and turbulence—dominate over additive noise. Interference-aware SIMO processing is therefore required to maintain throughput in compact OAM-based optical interconnects.

Conclusion

We experimentally demonstrate a compact VCSEL–metasurface OAM-based optical wireless link employing modal diversity in a SIMO architecture for short-reach interconnects, achieving multi-Gbps transmission under both static and weak-turbulence conditions. At the HD-FEC threshold, ZF improves throughput by up to 20% over MRC without turbulence and 24% under weak turbulence ($D/r_0 = 0.2$). This work focuses on system-level performance under practical receiver constraints rather than ideal mode-selective detection.

System performance is primarily limited by imperfect mode decomposition, branch correlation, and inter-branch mixing arising from hardware imperfections and turbulence, while achievable data rates are constrained by APD bandwidth. These can be improved using higher-bandwidth or fiber-coupled receivers, while replacing SLM-based mode decomposition with metasurface implementations would enable a compact, power-efficient receiver.

Overall, compact OAM-based interconnects require interference-aware SIMO processing to mitigate inter-branch mixing and branch correlation and achieve robust high-throughput operation in intra-data-center environments.

Acknowledgements

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